INCH-POUND
MIL-DTL-24643/19G
27 November 2023
SUPERSEDING
MIL-DTL-24643/19F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 1,000 VOLTS, TYPE LS6SGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight with Circuit Integrity)

First - Copper conductor, coated or uncoated (see table I for size).

Second - Thermoset insulation, extruded or taped (see table I for thickness). Standard identification code applied by method 1 or 3, or letter identification code applied by method 5, with two conductors each A (or black), B (or white), and C (or red). If identification method 1 is used, outer surface to be printed shall appear black.

Third - The six conductors shall be cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be A, B, C, A, B, C (or black, white, red, black, white, red). A single conductor shall not be used as the central core for the cable. Fillers may be used as necessary to form a firm, well-rounded assembly.

Fourth - An optional binder.

Fifth - Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/19	Type and size	Conductor size (AWG)	Cable overall diameter max. (inches)	Insulation thickness min. avg. (inch)	Jacket thickness min. avg. (inch)	Conductor resistance max. (ohms)	Insulation resistance (megohms)	Accelerated service loading (amperes)
-01UN	LS6SGU-100	0 (class D)	1.600	0.035	0.060	0.1060	160	475
-02UN	LS6SGU-125	00 (class D)	1.790	0.040	0.060	0.0888	150	520
-03UN	LS6SGU-150	000 (class D)	1.960	0.040	0.060	0.0674	135	570
-04UN	LS6SGU-200	0000 (class D)	2.200	0.050	0.075	0.0530	125	650

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	5,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	See table I
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation (extruded insulation only)	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±34 °F [95±1 °C]) (see MIL-DTL-24643 for limits of drip allowance)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60

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Elongation (percent of unaged, min.)	60
Permanence of printing (insulation extruded coverings only) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure

Group D:

Gas flame (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (-4±3.6 °F [-20±2 °C])	No damage
Gas flame (circuit integrity) (3 hours)	No failure
Accelerated service (see table I for load current)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	

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Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption (extruded insulation only)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS6SGU all sizes	800

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-035)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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